

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– VI (NEW) EXAMINATION – WINTER 2021****Subject Code:3161013****Date:30/11/2021****Subject Name:Systems Engineering****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) What is system engineering? Give examples of systems requiring system engineering.	03
	(b) What is the importance of risk management?	04
	(c) What is the purpose of WBS in project management?	07
Q.2	(a) What is critical path method (CPM) in project management?	03
	(b) What is SEMP in systems engineering?	04
	(c) Explain principal stages in a system life cycle.	07
	OR	
	(c) What are systems engineering methods? Explain in detail.	07
Q.3	(a) What is meant by "MOE" and "MOP"?	03
	(b) Describe Needs analysis phase in the system life cycle.	04
	(c) Explain concept exploration phase in the system life cycle.	07
	OR	
Q.3	(a) What is meant by the term “ functional allocation ”?	03
	(b) Differentiate the terms "concept selection" and "concept validation".	04
	(c) Write a short note on Model Based Systems Engineering (MBSE).	07
Q.4	(a) What is the purpose of the test and evaluation master plan?	03
	(b) Explain the concept of risk reduction.	04
	(c) Explain in detail prototype development as a risk mitigation techniques.	07
	OR	
Q.4	(a) Define the terms Preliminary Design Review (PDR).	03
	(b) Explain the Concepts of maintainability and availability for a system.	04
	(c) Explain the concept of User Interface Design (UID).	07
Q.5	(a) Define the terms Critical Design Review (CDR).	03
	(b) Explain the Concepts of reliability and redundancy for a system.	04
	(c) Explain the concept of system integration.	07
	OR	
Q.5	(a) What is operational readiness testing in a system?	03
	(b) What is hypothesis testing in engineering?	04
	(c) Explain the concept of system upgrades and modernization.	07
